

MCP Context Block Spec

This document describes how outside LLMs read and write LYKN's synthesis layer through the Model Context Protocol (MCP). LYKN exposes a Streamable HTTP MCP server authenticated via OAuth or personal access token.

Session bootstrap

Every new conversation MUST call `lykn_getContextBlock` before responding to the user. Do not wait for the user to mention LYKN — they connected this server precisely so context loads automatically.

Context block contents

- Beliefs — ratified principles with IDs; models should treat these as governance.
- Rules — if-then policies; record `lykn_recordRuleApplication` when one is followed.
- Facts — confirmed truths about the user; prefer these over model guesses.
- Project state — key/value pairs for the active project (goal, status, blockers).

Scoped read tools

Use `lykn_getBeliefs`, `lykn_getRules`, `lykn_getFacts`, `lykn_getProjectState`, `lykn_searchVault`, and `lykn_getContextBlock` only when mid-conversation scoped access is needed. For the upfront load, `getContextBlock` alone is sufficient.

Write-back tools

- `lykn_proposeBelief` — propose a new belief for user ratification; never auto-ratify.
- `lykn_pushProjectState` — persist durable project decisions for the next session.
- `lykn_recordRuleApplication` — audit when a returned `rule_id` was actually applied.

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Supported clients

- Claude Code — ``claude mcp add`` with OAuth; confirm via ``claude mcp list``.
- Cursor — Settings! MCP; Streamable HTTP with MCP-OAuth.
- ChatGPT — custom MCP connector (Pro/Team/Enterprise + Developer Mode).
- Gemini CLI — ``gemini mcp add``; synthesis tools available in the CLI agent.
- LYKN in-app chat — same synthesis layer, native integration.

Security

Tokens are revocable from Connections. OAuth consent shows exactly which synthesis scopes a client receives. Rate limits and audit logging apply to tool calls; suspicious patterns surface in the security audit trail.

Example: project state push

```
lykn_pushProjectState({ project_id: "lykn-mcp", state_key: "transport", state_value: "Streamable HTTP MCP, hand-rolled core" })
```

The next AI client that loads context for that project picks up the decision without the user re-explaining it.